Aaryan Samanta

Cupertino, CA | aaryan.samanta@gmail.com | github.com/aaryansamanta
Google Scholar | ORCID 0009-0005-6381-4890 | aiethos.org | nextgenai.org

High school researcher applying ML to biology and medicine, with two peer-reviewed papers, USACO Platinum, USAPhO Gold, and founder of AI Ethos, Inc., a 501(c)(3) providing multilingual AI tutoring to underserved students.

EDUCATION

Legend College Preparatory, Cupertino, CA — Class of 2028

- GPA 4.92 weighted; SAT 1570 (800 Math, 770 EBRW); Track: Computer Science and Engineering with AI and Technology
- AP Scholar with Distinction: 9 exams, 4.77 average; scores of 5 in Calculus BC, Physics C (Mechanics and E&M), Physics 1, Physics 2 and Precalculus
- College Board National Recognition Program (Outstanding Academic Achievement); NSHSS National Scholar

RESEARCH AND PUBLICATIONS

Sole Author, IEEE AIAM 2025 — Nov 2025

Quantum-Inspired Hybrid Genetic Algorithm and Graph Neural Network Ensemble for Multimodal Classification

- Fused a quantum-inspired genetic algorithm with a graph neural network, interpreted with SHAP; F1 0.57 vs. 0.55 (Random Forest) and 0.52 (SVM). Paper (IEEE Xplore) | Code and data

First Author, International Journal of High School Research, Vol. 8, Issue 16 — Aug 2026

Natural Genetic Variation in Mitochondrial Health-Regulating Genes in the C. elegans Strains CX11314 and EG4725 Leads to Increased Mitochondrial Resilience

- Analyzed 540 isotypes and 609 RNA-seq samples across six mitochondrial quality-control genes; built an expression predictor reaching Spearman 0.959. Paper (DOI) | Code and data

Independent Mentored Researcher, Stanford Medicine, Department of Urology — Jan–Jun 2026

Mentor: Christos E. Constantinou, Associate Professor of Urology, Emeritus

- Analyzed anonymized pelvic-floor ultrasound from 23 subjects, extracting displacement, velocity and acceleration during voluntary contraction, Valsalva and cough. Preliminary finding: voluntary contraction increases urethral support; cough produced no significant caudal displacement. Project notes

Independent Mentored Researcher, MIT EECS — Oct 2026–present

Mentor: Prof. Robert Berwick

- Conduct weekly research applying machine learning and data analysis to computational linguistics

Student Researcher, UCLA COSMOS, Cluster 1: Brain-Inspired Computing — Summer 2026

- Co-developed and presented (team poster) InfernoCommand, an actor-critic, fire-relative reinforcement learning system for wildfire response, built on a real Los Angeles terrain model (SRTM elevation, OpenStreetMap, NOAA weather, January 2025 Palisades Fire). Over training, mean reward rose 52% and buildings lost fell 54%; manuscript in progress. Code

Student Researcher, UCSB Summer Research Academies, Track 5: Molecular Clock — Summer 2025

- Team study of natural genetic variation in mitochondrial-regulating genes in C. elegans; presented at the Capstone Seminar, July 24, 2025

AI Intern (2025) and DS/ML Capstone Student (2025–2026), Legend College Preparatory — 2025–2026

- Completed a three-course AI/ML capstone: CNN pneumonia detector, symptom and disease pattern analysis, and diabetes-risk model comparison. Code

HONORS AND AWARDS

- **USACO Platinum Division (2026):** perfect scores (1000/1000) in Bronze, Silver and Gold; ranked #1 among all pre-college participants in each division
- **USAPhO Gold Medal (2026):** F=ma score of 24/25, second-highest in the U.S.
- **USABO Semifinalist (2026)**
- **AMC:** perfect scores on AMC 8 (25/25) and AMC 10 (150/150) in 2024; AIME qualifier
- **Kaggle Silver Medal (2026):** Stanford RNA 3D Folding Part 2, 92nd of 1,867 teams (top 4.9%).
- **ARML Local (2026), Team WARML I-5:** 2nd globally and 1st among U.S. teams, among 300 teams and 2,000+ competitors. Official results

LEADERSHIP AND SERVICE

Founder, President and CEO, AI Ethos, Inc. (501(c)(3)) | aiethos.org 2025–present

- Built multilingual, privacy-first AI tutoring for low-income, rural, multilingual and neurodiverse students; reached 75+ students in three languages with an average 22% gain in math scores
- Received IRS approval in September 2025; launched a program for San Jose's Spanish-speaking community in June 2026

Founder and President, NextGenAI International | nextgenai.org 2025–present

- Lead a student-run AI education organization offering bootcamps, mentored projects and exchange; led Taiwan and Thailand roadshows and signed a university MOU; 100+ participants across three countries

Co-captain and Student Coach, Cupertino Junior Olympiad Math/CS Team 2024–present

- Coached 20+ peers across 5+ districts toward AIME qualification and USACO Gold

Volunteer Team Leader, VA Palo Alto Health Care System 2024–present

- Led a student team that reviewed 25+ patient medical images under staff supervision and developed a standardized review workflow to support clinical image access

Team Lead and Mentor, AP Calculus Tutoring and Faria STEM Outreach Ongoing

- Tutored 15+ peers in multivariable calculus and graph theory; mentored 20+ elementary students in coding

Founder and Research Lead, CogArc In development

- Wearable and EEG neurofeedback platform testing whether emotion is encoded in how body signals couple (hypothesis stage)

BEYOND THE CLASSROOM

Track and Field, Monta Vista Matadors 2024–present

- Freshman-year school record in the 100 m (11.89); Rookie of the Year. Anchored the winning frosh/soph 4×400 m relay at the Los Gatos Top 8 Invitational (3:36.02); ran on the 4×400 m relay that set an all-time stadium record at RustBuster
- 2026 season bests: 100 m 11.85, 200 m 24.35, 400 m 54.53. Athletic.net

Elected Percussion Section Leader, Monta Vista Concert Band 2025

- Directed rehearsals leading to the first-ever unanimous Superior rating across all Monta Vista ensembles at the CMEA Festival

SKILLS

Languages and tools: Python (PyTorch, TensorFlow, scikit-learn, NumPy, Pandas), Java, JavaScript, MATLAB, Git, LaTeX, Linux
Spoken: English; Spanish (AP level); Bengali (beginner)